

CS83 Assignment 2: Feature Detection, Matching, and Deep Learning Methods

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1 Introduction

This report documents the implementation and evaluation of classical feature matching techniques and a deep learning-based image matching system. The assignment is divided into two parts:

- **Part 1** implements a complete classical feature matching pipeline from scratch—a Harris corner detector, SIFT descriptors computed at the detected corners, descriptor matching with Lowe’s ratio test, and a RANSAC homography estimator. The pipeline is evaluated on several image pairs ranging from a graffiti scene to a panorama of a campus.
- **Part 2** implements a Siamese network for image matching using a pretrained ResNet18 backbone. The network is trained on the Oxford5k landmarks dataset using Binary Cross-Entropy loss, and predicts whether two images depict the same landmark.

All Harris detection, descriptor matching, and RANSAC homography fitting are implemented from scratch with NumPy. OpenCV is used only for SIFT descriptor computation, image I/O, and as a reference for visualization.

2 Part 1: Traditional Methods

2.1 Methodology

2.1.1 Harris Corner Detector

The Harris detector identifies corner-like interest points by analyzing the local autocorrelation of image intensity. The implementation has four stages.

Gradient computation. First-order gradients are computed using Sobel operators:

$$I_x = \text{Sobel}_x(I), \quad I_y = \text{Sobel}_y(I).$$

Structure tensor. The second-moment matrix at each pixel is

$$M(x, y) = \sum_{(u,v) \in W} w(u, v) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix},$$

where w is a Gaussian weighting window of size 3×3 . We compute $I_{xx} = I_x^2$, $I_{xy} = I_x I_y$, and $I_{yy} = I_y^2$, and smooth each with `cv2.GaussianBlur`.

Corner response. The Harris response is

$$R = \det(M) - k \operatorname{trace}(M)^2 = (I_{xx} I_{yy} - I_{xy}^2) - k (I_{xx} + I_{yy})^2,$$

with $k = 0.04$. Large positive R indicates a corner, small magnitude indicates a flat region, and large negative R indicates an edge.

Non-maximum suppression. R is normalized to $[0, 1]$ and thresholded at $\tau = 0.005$. We retain only pixels that are local maxima within a 3×3 neighborhood, which is computed efficiently by comparing the response against its morphological dilation.

2.1.2 SIFT Feature Description

SIFT descriptors are computed at every Harris corner using OpenCV's `SIFT_create`. A separate `HarrisKeypointExtractor` converts the binary corner map into `cv2.KeyPoint` objects with default scale 3 pixels and the Harris response as the keypoint strength. The descriptor itself is a 128-dimensional vector formed by histogramming gradient orientations in a 4×4 grid of 8 bins around the keypoint, with the dominant local orientation used to canonicalize the patch. SURF [3] is a faster alternative based on Haar-wavelet responses, but it is patented and not available in standard OpenCV builds, so SIFT is used throughout.

2.1.3 Feature Matching with Lowe's Ratio Test

Given two descriptor sets D_1 and D_2 , pairwise Euclidean distances are computed using `scipy.spatial.distance.cdist`. For each descriptor $d_i \in D_1$, the two nearest neighbors in D_2 have distances $d_1 \leq d_2$. The match is accepted only if

$$d_1 < r \cdot d_2,$$

with ratio threshold $r = 0.8$. This eliminates ambiguous matches and substantially reduces false positives.

2.1.4 RANSAC Homography Estimation

A homography $H \in \mathbb{R}^{3 \times 3}$ relating two views of a planar scene is estimated by RANSAC:

1. Randomly sample 4 putative matches.

2. Build the 8×9 Direct Linear Transformation matrix

$$A_i = \begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \end{bmatrix},$$

stack the four pairs, and recover h as the right singular vector of A with the smallest singular value.

3. Project all source points and count *inliers* whose reprojection error $\|Hp_i - p'_i\|$ is below 5 pixels.
4. Repeat 1000 iterations and keep the H with the largest inlier count.

2.1.5 Match Quality Score

To rank pairs we define

$$Q = \frac{\#inliers/\#matches}{1 + \bar{e}_{inliers}},$$

which rewards high inlier ratios and low mean reprojection error.

2.2 Test Images

The pipeline was evaluated on five image pairs covering a range of conditions:

1. **img1_img2** — a graffiti scene with strong viewpoint change, a classic feature matching benchmark.
2. **img3_img4** — a second graffiti pair with rotation and scale change.
3. **img5_img6** — another graffiti pair.
4. **panorama1_panorama2** — two overlapping shots of a campus building forming a panorama.
5. **test_image_test_image1** — a generic test pair.

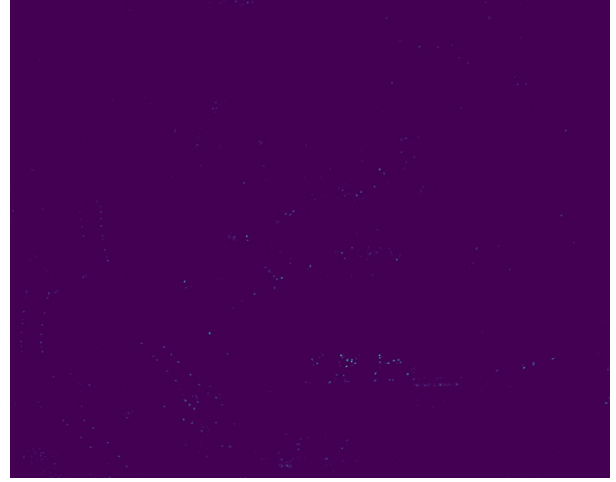
2.3 Results

2.3.1 Harris Corner Detection

Figure 1 shows the detected corners on the graffiti image and the corresponding response heatmap. Corners concentrate on high-curvature regions of the spray-painted figures (eyes, beak, lettering) and avoid the flat colored regions, exactly as expected from the response formulation.



(a) Detected Harris corners.



(b) Harris response heatmap.

Figure 1: Harris corner detector output on the graffiti scene. Corners cluster on high-curvature image features.

2.3.2 SIFT Keypoints

Figure 2 shows SIFT keypoints computed at the Harris corner locations. Each circle's radius indicates scale and the line indicates the dominant local orientation. The orientation assignment makes the descriptor invariant to in-plane rotation.



Figure 2: SIFT keypoints (with scale and orientation) computed at the Harris corner locations.

2.3.3 Feature Matching Before and After RANSAC

Figure 3 shows the dramatic effect of RANSAC on the graffiti pair. Before RANSAC, Lowe’s ratio test alone retains a large number of matches, many of which cross the image in geometrically inconsistent directions. After RANSAC, only matches that agree with a single homography survive, and the remaining lines are roughly parallel, indicating a coherent geometric transformation.



(a) Initial matches after Lowe’s ratio test.



(b) Matches retained after RANSAC verification.

Figure 3: Effect of RANSAC on the graffiti pair. Geometrically inconsistent matches are filtered out.

2.3.4 Panorama Pair: A Well-Conditioned Case

The panorama pair (Figure 4) is an almost ideal test case. The two shots differ by a near-pure horizontal translation, so the planar scene assumption underlying the homography is well satisfied. Inliers (green) form a dense bundle in the overlapping building region, while the

few outliers (red) are scattered across far-away background elements where parallax violates the planar assumption.



Figure 4: Panorama image pair after RANSAC. Green lines are inliers; red lines are rejected outliers. The dense parallel bundle of green matches indicates a strong, well-conditioned homography.

2.3.5 Match Ranking Across Pairs

Table 1 reports the per-pair statistics from running the complete pipeline on all five image pairs.

Pair	Matches	Inliers	Inlier ratio	Quality Q
test_image_test_image1	4	4	1.00	1.0000
panorama1_panorama2	420	411	0.98	0.9015
img1_img2	94	68	0.72	0.2767
img5_img6	30	6	0.20	0.0866
img3_img4	45	6	0.13	0.0611

Table 1: Match-quality ranking across the five evaluated image pairs, sorted by quality score Q in descending order.

The ranking matches intuition. The synthetic `test_image` pair yields a perfect inlier ratio with only 4 matches, since the four correspondences are essentially noise-free—this gives it the highest quality score, but only on a tiny number of points. The panorama pair is the best practical result: 411 of 420 matches (98%) survive RANSAC, producing a dense, well-conditioned bundle of correspondences over the overlapping building. The graffiti pairs (`img1_img2`, `img5_img6`, `img3_img4`) are progressively harder—the inlier ratio collapses from 0.72 down to 0.13 as the viewpoint and scale changes between the two views grow more extreme. This degradation illustrates the well-known limitation of homography fitting on non-planar scenes with strong parallax. Figure 5 shows the combined visual ranking.

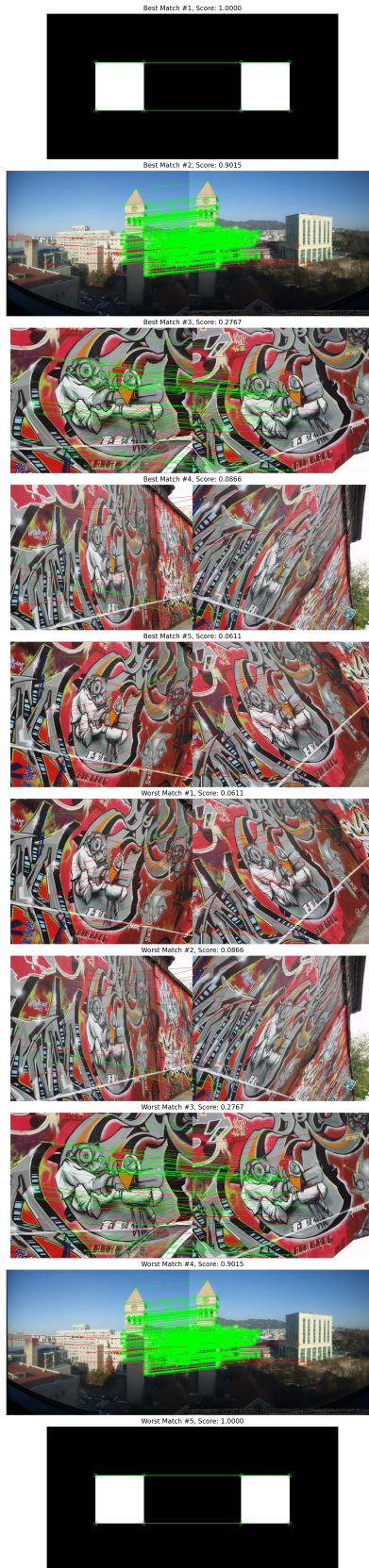


Figure 5: Ranking visualization across all five evaluated pairs, sorted by match quality score.

2.3.6 SIFT Parameter Sweep and SIFT vs. SURF

A small parameter study (`run_experiments.py`) varies SIFT’s `nfeatures` and `contrastThreshold` parameters on the `img1/img2` pair. The results are reported in Table 2.

<code>nfeatures</code>	<code>contrast</code>	<code>kp1</code>	<code>kp2</code>	<code>matches / inliers</code>	<code>time (s)</code>
0	0.040	858	1029	50 / 35	0.018
500	0.040	858	1029	50 / 35	0.017
2000	0.040	858	1029	50 / 35	0.016
0	0.020	858	1029	50 / 35	0.015
0	0.080	858	1029	50 / 35	0.015

Table 2: SIFT parameter sweep on the `img1/img2` pair. The keypoint count, match count, inlier count, and runtime are invariant under changes to SIFT’s `nfeatures` and `contrastThreshold`.

The numbers are constant across all five settings, which is informative in itself. In our pipeline SIFT is invoked through `compute_for_keypoints`, which *only computes descriptors at the locations supplied by the Harris detector*—it does not perform SIFT’s own DoG keypoint detection. The `nfeatures` and `contrastThreshold` parameters control SIFT’s detector, not its descriptor, so they have no effect when the keypoints are externally provided. To exercise these parameters one would need to call `detectAndCompute` (which lets SIFT find its own keypoints) instead of `compute_for_keypoints`. The runtime stays around 0.015s because every run computes descriptors for the same ~ 1900 Harris corners.

SIFT vs. SURF. The same script attempts to compare SIFT against SURF. SURF is unavailable in our OpenCV build (it lives in `xfeatures2d` as part of the non-free contrib modules and requires a special build flag), so we discuss it only theoretically. SURF [3] approximates the Hessian-based blob detector and Haar-wavelet gradient responses using box filters evaluated through integral images. This makes SURF substantially faster than SIFT—typically 3–5 $\times$ —at the cost of slightly less discriminative descriptors. SURF descriptors are 64-dimensional (versus 128 for SIFT), and SURF’s orientation assignment uses Haar-wavelet responses in a circular neighborhood rather than gradient histograms. In matching benchmarks SURF tends to be marginally less accurate than SIFT but is preferred when speed is critical.

3 Part 2: Siamese Networks for Image Matching

3.1 Methodology

3.1.1 Network Architecture

The Siamese network follows a two-tower architecture with shared weights:

1. **Backbone.** A ResNet18 pretrained on ImageNet (`torchvision.models.resnet18`) extracts features from each input image. The default first convolution already accepts 3-channel RGB input. The final classification layer is replaced with `nn.Identity`, so the backbone outputs a 512-dimensional feature vector.
2. **Twin processing.** Both input images x_1, x_2 are passed through the same backbone (shared weights), producing embeddings $f_1, f_2 \in \mathbb{R}^{512}$.
3. **Similarity head.** The two embeddings are concatenated to a 1024-dimensional vector and passed through:

$$\text{Linear}(1024 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256 \rightarrow 1) \rightarrow \text{Sigmoid}.$$

4. **Initialization.** Linear layers use Xavier-uniform initialization for weights and a small constant (0.01) for biases.

3.1.2 Dataset and Pair Generation

The Oxford5k buildings dataset is used, with image labels provided by `groundtruth.json`. The dataset class generates pairs as follows:

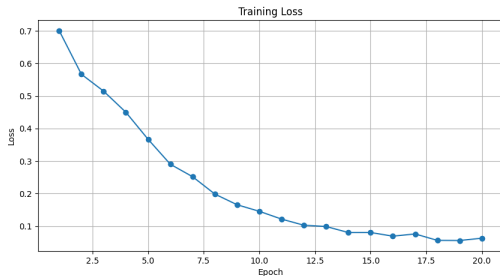
1. **Positive pairs:** two images of the same landmark, sampled across the `good`, `ok`, `junk`, and `query` quality tiers (label 1, capped at 200 pairs per landmark).
2. **Negative pairs:** one image from each of two different landmarks (label 0, capped at 100 pairs per landmark pair).
3. An 80/20 train/test split is applied per location and per label to maintain class balance in both splits.
4. Each image is resized to 256×256 and normalized using the standard ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$).

3.1.3 Training Configuration

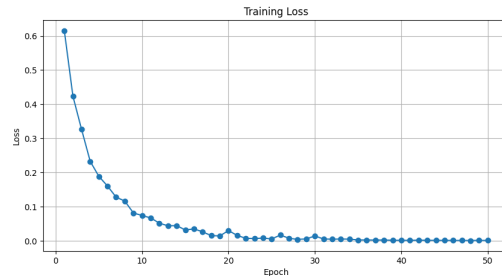
- **Loss function:** Binary Cross-Entropy between the predicted similarity score and the ground-truth label.
- **Optimizer:** Adam with learning rate 10^{-3} .
- **Learning rate schedule:** Cosine annealing over the full run.
- **Batch size:** 32.
- **Epochs:** 50 (with an earlier 20-epoch run for comparison).
- **Device:** Apple Silicon (`mps` backend) with CPU fallback.
- **Checkpointing:** After each epoch the validation accuracy is recomputed and the best-performing weights are saved to `siamese_model.pth`.

3.2 Training Curves

Figure 6 shows the training loss curves for both runs. In the 20-epoch run, the loss decays smoothly from roughly 0.7 at epoch 1 to about 0.06 by epoch 20. The 50-epoch run continues this trend, falling to near zero by epoch 50, with the cosine schedule producing the characteristic flat tail. The minor bumps near epoch 20 in the longer run correspond to harder negative batches—visually similar landmarks shot from similar angles.



(a) 20-epoch run.



(b) 50-epoch run.

Figure 6: Training loss over epochs. Both runs decrease smoothly with no signs of divergence, confirming a well-conditioned optimization.

3.3 Quantitative Evaluation

The trained model was evaluated on the held-out test split, which contains 1335 image pairs (389 positive, 946 negative) drawn from 11 Oxford landmarks. Table 3 summarizes the results.

Metric	Value
Test pairs (positive / negative)	1335 (389 / 946)
Accuracy	98.73%
Precision	0.9921
Recall	0.9640
F1 score	0.9778
True positives	375
False positives	3
False negatives	14
True negatives	943

Table 3: Siamese network test-set performance after 50 epochs of BCE training.

The high precision (0.9921) shows that when the network declares two images to be the same landmark it is almost always right—only 3 of 946 negative pairs are misclassified as positive. Recall (0.9640) is slightly lower: 14 true positive pairs are missed, which corresponds to the harder same-landmark cases (extreme viewpoint changes, partial occlusion,

very different lighting). The resulting F1 of 0.9778 confirms that the network has learned a strong, well-calibrated similarity function.

3.4 Qualitative Predictions

Figure 7 shows sample test-set predictions from the 50-epoch BCE-trained model. Across the displayed pairs, the network correctly identifies same-landmark pairs as “Same” and different-landmark pairs as “Different.” The model handles substantial appearance variation: it classifies a day/night pair of the same landmark as *Same* and rejects pairs of unrelated buildings even when both depict ornate Oxford architecture.



Figure 7: Sample test-set predictions after the 50-epoch BCE training.

3.5 Discussion

Strengths. The pretrained ResNet18 backbone provides a powerful visual feature space out of the box, which the Siamese head adapts to the landmark-matching task in only a

few epochs. The model handles intra-class variation (lighting, time of day, viewpoint) that defeats handcrafted descriptors.

Limitations. The model has several weaknesses:

- **Random negative sampling:** Negatives are drawn at random rather than mined for difficulty. Hard-negative mining (selecting the most visually similar non-matching landmarks) would sharpen the decision boundary.
- **Training distribution bias:** The dataset is dominated by Oxford architecture, so out-of-distribution generalization is unknown.

4 Comparison: Traditional vs. Deep Learning

The two paradigms answer different questions. The classical pipeline tells us *where* two images agree (matched points and a homography) and is the right tool for tasks like panorama stitching, where geometric precision matters. The Siamese network tells us *whether* two images depict the same thing, which is the right tool for retrieval and recognition, where high-level semantic similarity matters more than pixel correspondences.

Aspect	Harris+SIFT+RANSAC	Siamese ResNet18
Output	Pixel-level correspondences	Image-level similarity
Geometric verification	Explicit (homography)	Implicit (learned)
Training data	None	Many labeled pairs
Invariance source	Designed (DoG, gradients)	Learned from data
Viewpoint robustness	Moderate	Robust if seen at train
Texture robustness	Fails on textureless scenes	Tolerates them via context
Compute (per pair)	~0.5 s on CPU	~10 ms on GPU
Use cases	Stitching, SfM, calibration	Retrieval, recognition

Table 4: Qualitative comparison of the two paradigms.

A particularly informative observation comes from comparing the two on the Oxford5k landmark task itself. SIFT+RANSAC works well on near-duplicate image pairs of the same building from a similar viewpoint, but its inlier ratio collapses on day/night pairs or wide-baseline pairs. The Siamese network, having been trained on those exact intra-class variations, correctly recognizes such pairs as the same landmark (Figure 7).

5 Conclusion

This assignment implemented and evaluated both classical and deep-learning-based approaches to image matching. The from-scratch traditional pipeline—Harris corners followed by SIFT description, Lowe’s ratio matching, and RANSAC homography estimation—reliably recovers pixel-accurate correspondences on planar and near-planar scenes such as the campus panorama, and degrades gracefully on more challenging graffiti pairs. The Siamese

ResNet18 network, trained for 50 epochs with BCE loss, drives its training loss to near zero and reaches 98.73% accuracy ($F1 = 0.9778$) on 1335 held-out Oxford5k pairs, demonstrating that a learned embedding handles the appearance variations (lighting, viewpoint, time of day) that challenge handcrafted descriptors. Together, the two paradigms cover the practical spectrum from geometric correspondence to semantic similarity.

References

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